

8(a)	<i>any three from</i> 1 detect / respond to, (change in) stimulus / stimuli ; 2 <i>two examples from</i> – light / heat / sound / touch / pressure / pain / chemicals / taste / smell / tension ;; 3 (act as) transducers / convert stimulus energy to electrical energy ; 4 produce, generator / receptor / action, <u>potential</u> ; 5 passes impulse, to / along, <u>sensory</u> neurone ;	3
8(b)(i)	<i>Na⁺/K⁺ pump: C or D and E ;</i> <i>Na⁺ channels open: C ;</i> <i>K⁺ channels open: D ;</i>	3
8(b)(ii)	<i>for A (ora for B)</i> 1 <u>generator</u> / <u>receptor</u>, <u>potential</u> ; 2 <u>threshold</u> not, reached / exceeded / crossed ;	2

Question	Answer	Marks
8(b)(iii)	<i>any two from</i> 1 limits / controls, (maximum) frequency of action potentials ; 2 (action potentials / impulses) travel in one direction ;	2
8(b)(iv)	<i>any four from</i> 1 local circuit / movement of ions from positive to negative region ; 2 (causes) opening of Na⁺ channels ; 3 at next, node (of Ranvier) / gap in myelin sheath ; 4 causing next / new, action potential / depolarisation ; 5 saltatory conduction ; 6 one-way transmission ; 7 AVP ; e.g. myelin (sheath) / Schwann cells, insulate axon / prevent (named) ion movement / impermeable to (named) ions / speed up transmission / lengthens local circuits	4